Structurally, these early drones were designed with simple fixed-wing configurations, typically made from lightweight materials such as wood, fabric, and early aluminum alloys. Propulsion was achieved using small piston engines or early internal combustion systems. Aerodynamic efficiency was minimal but sufficient for short-range, low-altitude missions. The designs were often modeled after small conventional aircraft but stripped of cockpit space to accommodate control systems and fuel.

One of the earliest prototypes, the Aerial Target (1916), developed by A. M. Low in the United Kingdom, demonstrated the first attempt at creating a pilotless aircraft controlled via radio waves. It was followed by the Queen Bee (1935), a radio-controlled target drone developed by the Royal Navy and Royal Air Force for anti-aircraft gunnery practice. The term “drone” itself is believed to have originated from this aircraft, named after the male bee known for its buzzing sound, similar to that produced by the aircraft’s engine. These pioneering platforms provided valuable insight into remote flight mechanics and control stability, setting the stage for future unmanned systems (Pliatsios, et al. 2021).

The operational use of first-generation drones was largely confined to military applications, including target towing, artillery training, and limited reconnaissance missions. Although their navigation capabilities were extremely primitive and their endurance limited, they provided defense forces with an invaluable tool for simulation and experimentation. These early UAVs also sparked growing interest among engineers and researchers, demonstrating that remote flight control could evolve into a viable and strategic technology.

Despite their technological simplicity, the first generation of drones played a critical role in establishing the conceptual and technical framework for modern UAV design. They introduced the principles of remote